Experiment 12: Decision Tree Classifier

Aim:

To build a Decision Tree Classifier that approves or rejects loan applications based on customer information such as age, income, credit score and existing loans.

Algorithm:

1. Prepare the dataset of customer records with the label Approve (1) or Reject (0).
2. Split the data into training and testing sets.
3. Select the attribute with the highest information gain (entropy criterion) as the root node.
4. Split the data on that attribute and repeat recursively for each branch until the maximum depth or a pure leaf is reached.
5. Print the learned tree rules and test the model on the test set to find its accuracy.
6. Predict the decision for a new applicant using the trained tree.

Code:

```
from sklearn.tree import DecisionTreeClassifier, export_text
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# Features: [Age, Income (in thousands), Credit Score, Existing Loans]
X = [
    [25, 30, 580, 2], [45, 80, 720, 0], [35, 55, 650, 1], [52, 95, 780, 0],
    [23, 20, 540, 3], [40, 70, 700, 1], [29, 35, 600, 2], [48, 90, 750, 0],
    [33, 45, 630, 1], [27, 52, 640, 3], [55, 100, 800, 0], [38, 60, 680, 1],
    [31, 65, 690, 2], [42, 85, 740, 0], [26, 32, 570, 2], [50, 75, 710, 1],
    [36, 58, 670, 1], [24, 22, 550, 3], [46, 38, 590, 2], [30, 42, 620, 2],
]
y = [0, 1, 1, 1, 0, 1, 0, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0, 0, 0]  # 1 = Approve, 0 = Reject
features = ["Age", "Income", "CreditScore", "Loans"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_state=1)

model = DecisionTreeClassifier(criterion="entropy", max_depth=3, random_state=42)
model.fit(X_train, y_train)

print("Decision Tree Rules:")
print(export_text(model, feature_names=features))

pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, pred))

new_customer = [[34, 50, 640, 1]]
result = model.predict(new_customer)[0]
print("New applicant", new_customer[0], "->", "Approved" if result == 1 else "Rejected")
```

Sample Output:

Output

```
Decision Tree Rules:
|--- CreditScore <= 660.00
|   |--- Loans <= 1.50
|   |   |--- class: 1
|   |--- Loans >  1.50
|   |   |--- class: 0
|--- CreditScore >  660.00
|   |--- class: 1

Accuracy: 1.0
New applicant [34, 50, 640, 1] -> Approved
```

Result:

The decision tree learned simple rules: applicants with a credit score above 660 are approved, and those with a lower score are approved only when they hold at most one existing loan. The model classified the test data correctly and approved the new applicant.